Bayesian ARX Identification for Robust Control and Attack-Resilient Analysis

Uncertainty-aware system identification for weak-data settings

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Core message

Bayesian methods preserve the set of plants still consistent with limited data and propagate that uncertainty into poles, frequency response, Nyquist geometry, and closed-loop behavior.

Why this matters for critical-infrastructure modeling

- Small utilities often need dynamic models for threat analysis, resilience studies, and mitigation planning, but they rarely have the budget to build those models from scratch.
- In practice, one works with whatever telemetry, topology, and operating history already exist — often incomplete, noisy, and weakly exciting.
- A nominal least-squares model can still look crisp under those conditions, even when the data leave many dynamics unresolved.

Practical thesis

Instead of forcing one overconfident plant model, preserve the set of plants still consistent with the data and propagate that uncertainty into robust-control quantities.

For critical-infrastructure modeling pipelines, Bayesian identification can reveal where existing data are insufficient while still allowing downstream analysis to be performed with explicit confidence statements.

ARX model: what we are trying to identify

- We start with a discrete-time SISO ARX model because it is interpretable, easy to estimate, and directly useful for control analysis.

$$y_t = \phi_t^\top \theta + e_t, \qquad e_t \sim \mathcal{N}(0, \sigma^2)$$

$$\phi_t = [y_{t-1}, \ldots, y_{t-n_a}, u_{t-1}, \ldots, u_{t-n_b}]^\top$$

- The parameter vector θ collects the autoregressive and input coefficients.
- If we identify only one best-fit θ, we lose information about every other model that still explains the data reasonably well.

Prior and likelihood

- Bayesian identification treats the unknown coefficient vector as a random variable before data are observed.

$$\theta \sim \mathcal{N}(\mu_0, \Sigma_0)$$

$$\mathsf{y} \mid \theta, \Phi \sim \mathcal{N}(\Phi\theta, \sigma^2 \mathsf{I})$$

- Σ_0 is the prior covariance matrix. It encodes how uncertain we are about the coefficients before seeing the data.
- Diagonal entries of Σ_0 are coefficient variances; off-diagonal entries capture how uncertainties move together.
- In the isotropic prior used here, $\Sigma_0 = \alpha^2 \mathsf{I}$, so every coefficient direction starts with the same uncertainty scale.

Posterior derivation, step 1: multiply prior and likelihood

- Bayes' rule: posterior $\propto$ likelihood $\times$ prior. For Gaussian terms, that product stays Gaussian after collecting the quadratic terms in θ.

$$\mathrm{p}(\theta \mid \mathcal{D}) \propto \mathrm{p}(\mathrm{y} \mid \theta, \Phi)\, \mathrm{p}(\theta)$$

$$\log \mathrm{p}(\theta \mid \mathcal{D}) = -\frac{1}{2\sigma^2} \|\mathrm{y} - \Phi\theta\|^2 - \frac{1}{2}(\theta - \mu_0)^\top \Sigma_0^{-1}(\theta - \mu_0) + \mathrm{c}$$

- The first quadratic term rewards agreement with the observed data.
- The second term penalizes moving too far from the prior mean unless the data justify it.

Posterior derivation, step 2: read off the new mean and covariance

- Completing the square in θ gives the posterior covariance and posterior mean.

$$\Sigma_{\mathrm{N}}^{-1} = \Sigma_0^{-1} + \frac{1}{\sigma^2}\Phi^\top\Phi$$

$$\mu_{\mathrm{N}} = \Sigma_{\mathrm{N}}\left(\Sigma_0^{-1}\mu_0 + \frac{1}{\sigma^2}\Phi^\top \mathrm{y}\right)$$

- $\Phi^\top\Phi$ is the information contributed by the data. Strong excitation makes this term informative in many directions.
- Σ_{N} becomes small only in directions the data actually identify. Weakly excited directions remain uncertain.

Posterior predictive distribution: uncertainty in output space

- Once we have $\mathrm{p}(\theta \mid \mathcal{D})$, we predict future outputs by integrating over parameter uncertainty instead of plugging in only one estimate.

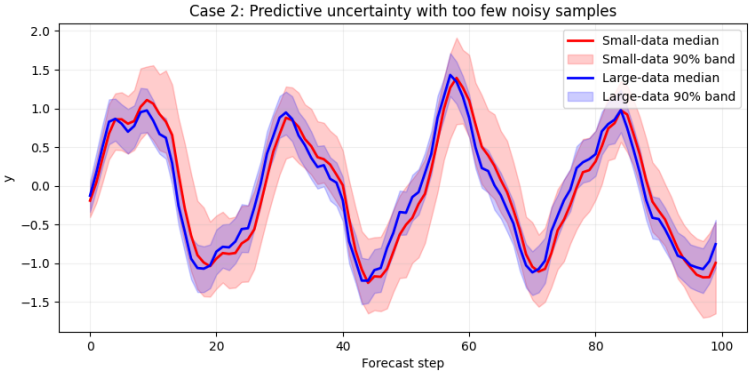
$$\mathrm{y}_\star \mid \mathcal{D}, \phi_\star \sim \mathcal{N}\Big(\phi_\star^\top \mu_\mathsf{N},\ \phi_\star^\top \Sigma_\mathsf{N} \phi_\star + \sigma^2\Big)$$

- The mean uses the posterior mean coefficients.
- The variance splits into parameter uncertainty ($\phi_\star^\top \Sigma_\mathsf{N} \phi_\star$) and observation noise (σ^2) (epistemic and aleatoric, respectively).

Controls interpretation

A predictive interval is an early warning that poles, margins, and closed-loop performance computed from one nominal plant may be fragile.

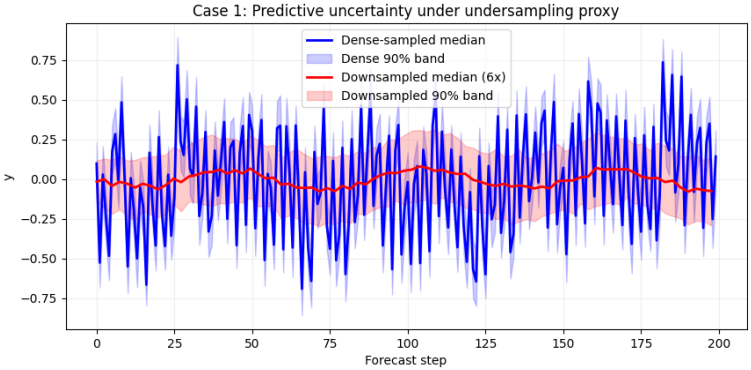
Too few samples: underconstrained identification



Interpretation

- The red small-data case uses too few noisy samples, so many parameter vectors remain plausible.
- The blue large-data case concentrates more tightly because the data provide more information.
- A sharp nominal fit can hide wide uncertainty in the induced dynamics.

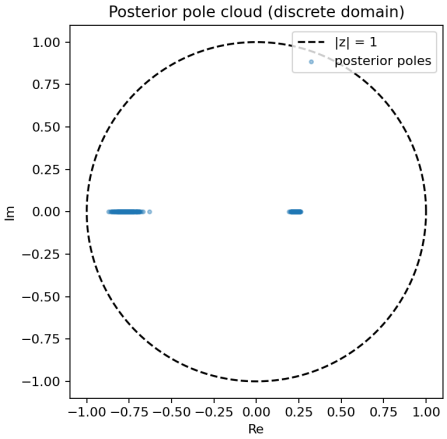
Undersampling proxy: what happens when the data rate is too coarse



Interpretation

- Sampling satisfying the Nyquist rate resolves fast dynamics; downsampling misses them.
- Coarse sampling lets multiple discrete-time models explain the same measurements.
- Bayesian identification exposes that unresolved ambiguity.

Posterior pole cloud: stability becomes probabilistic

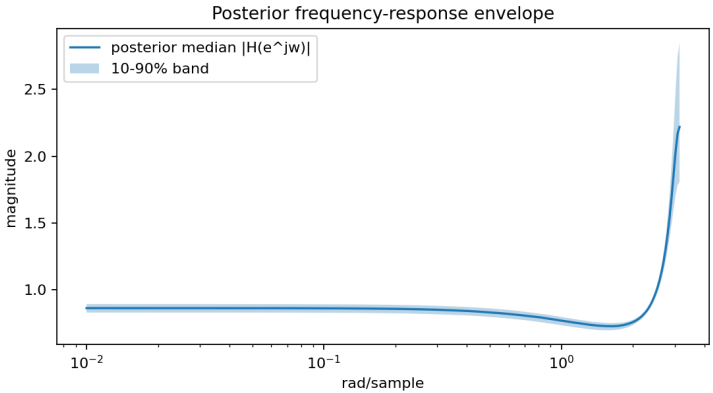


- Each point corresponds to poles obtained from one posterior sample of the identified ARX parameters.
- If posterior mass approaches the unit circle, a single stable nominal estimate is not persuasive evidence of robust stability.

$$\hat{p}_{\text{stab}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left\{\rho\left(A(\theta^{(i)})\right) < 1\right\}$$

Here $A(\theta^{(i)})$ is the discrete-time state matrix induced by posterior sample $\theta^{(i)}$.

Posterior frequency-response envelope

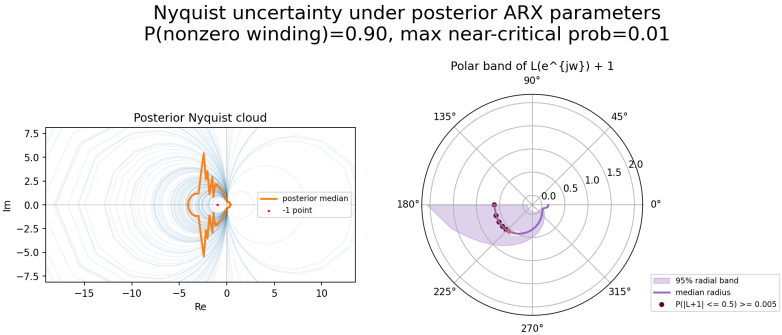


Interpretation

- The median curve is what a nominal design would inspect first.
- The shaded band shows where the posterior permits different frequency responses.
- Spread near crossover warns that nominal margins may be fragile.

This is not μ-analysis, but it is a useful empirical surrogate for epistemic uncertainty in robustness calculations.

Posterior Nyquist geometry



Interpretation

- The cloud comes from posterior plant samples.
- Proximity to the critical point -1 becomes probabilistic.
- Identification uncertainty maps directly into Nyquist robustness interpretation.

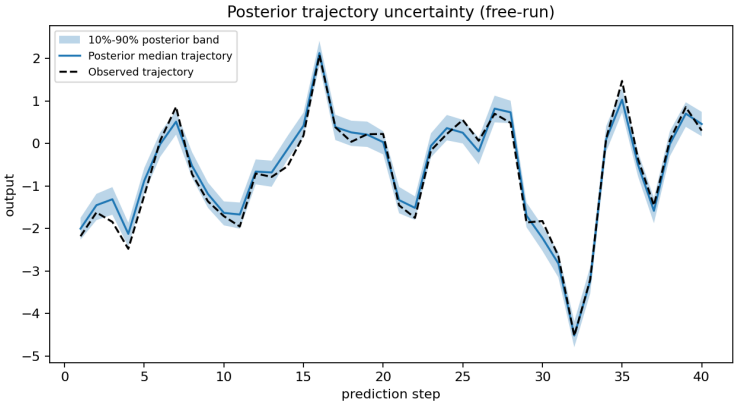
One-step-ahead versus recursive prediction

Interpretation

- One-step-ahead prediction uses the observed past output at each step.
- Recursive prediction feeds the model's own output back in.
- Their gap is a practical stress test for long-horizon reliability.

A model can look good one step ahead and still be fragile when rolled forward recursively.

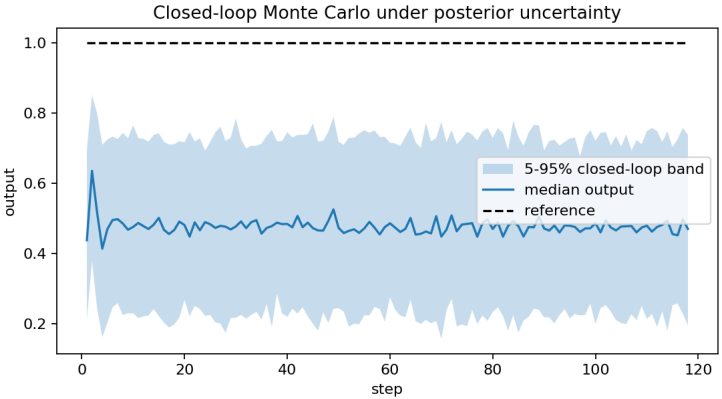
Posterior trajectory band under recursive propagation



Interpretation

- The band comes from repeatedly propagating posterior-consistent plants.
- Its width can expand or contract with the local dynamics.
- The figure exposes uncertainty from the model, not just measurement noise.

Closed-loop Monte Carlo: fix the controller, vary the plant



Interpretation

- Fix one controller, then sample many plants from $p(\theta \mid \mathcal{D})$.
- Simulate the closed loop for each sampled plant.
- The band reflects plant uncertainty, not repeated noise trials around one model.

If this band remains wide, nominal transient claims are not yet robust to identification uncertainty.

What this enables next: posterior-risk control and attack analysis

- Once the plant is a posterior distribution rather than one point estimate, design and adversarial analysis can be posed directly over that posterior.

$$\eta_c^\star \in \arg\min_{\eta_c} \mathrm{CVaR}_{1-\alpha}(\mathrm{J}_{\mathrm{cl}}(\eta_c, \theta)) \quad \text{s.t.} \quad \mathbb{P}_{\theta \sim \mathrm{p}(\theta|\mathcal{D})}(\rho(\mathrm{A}_{\mathrm{cl}}(\eta_c, \theta)) < 1) \geq 1 - \delta \tag{1}$$

- Posterior-risk controller design tunes against most plausible plants, not only the posterior mean.
- A matching attack-analysis problem is to look for perturbations that destabilize many posterior-consistent plants while remaining small in a visibility channel.
- This is the bridge from Bayesian system identification to Bayesian attack mitigation.

Takeaways

- Bayesian system identification is most valuable when the data are weak, underconstrained, or too coarsely sampled for a nominal model to be trusted on its own.
- The posterior maps directly into quantities control engineers already care about: poles, frequency response, Nyquist geometry, and closed-loop trajectories.
- For critical-infrastructure settings, we can still analyze models built from imperfect data while stating confidence explicitly instead of hiding uncertainty.
- The natural next step is to move from uncertainty analysis to posterior-aware design and mitigation.

Bottom line

Before trusting an identified plant in robust control or security analysis, ask what the posterior still permits.

Selected references

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